2.22 Hyperbolic Functions: Mix

Worked Solutions

1 a)

$$\begin{aligned}\det M &= \begin{vmatrix}\cosh x & \sinh x \\ \sinh x & \cosh x\end{vmatrix} \\ &= \cosh^2 x - \sinh^2 x = 1.\end{aligned}$$

As $\det M = 1$, the transformation T preserves area.

1 b)

$$\begin{pmatrix}\cosh x & \sinh x \\ \sinh x & \cosh x\end{pmatrix}\begin{pmatrix}0 & 1 & 0 \\ 0 & 0 & 1\end{pmatrix} = \begin{pmatrix}0 & \cosh x & \sinh x \\ 0 & \sinh x & \cosh x\end{pmatrix}.$$

So $O'(0,0)$, $A'(\cosh x, \sinh x)$ and $B'(\sinh x, \cosh x)$ are the images of O, A and B respectively.

$$\begin{aligned}O'A'^2 &= (\cosh x - 0)^2 + (\sinh x - 0)^2 = \cosh^2 x + \sinh^2 x, \\ O'B'^2 &= (\sinh x - 0)^2 + (\cosh x - 0)^2 = \sinh^2 x + \cosh^2 x, \\ A'B'^2 &= (\sinh x - \cosh x)^2 + (\cosh x - \sinh x)^2 \\ &= \sinh^2 x - 2\sinh x \cosh x + \cosh^2 x + \cosh^2 x - 2\sinh x \cosh x + \sinh^2 x \\ &= 2\sinh^2 x - 4\sinh x \cosh x + 2\cosh^2 x.\end{aligned}$$

The triangle is equilateral if $O'A' = O'B' = A'B'$:

$$\begin{aligned}2\sinh^2 x - 4\sinh x \cosh x + 2\cosh^2 x &= \cosh^2 x + \sinh^2 x \\ \Rightarrow \sinh^2 x + \cosh^2 x - 4\sinh x \cosh x &= 0 \\ \Rightarrow \cosh 2x - 2\sinh 2x &= 0 \\ \Rightarrow 1 - 2\tanh 2x &= 0 \\ \Rightarrow \tanh 2x &= \frac{1}{2} \\ \Rightarrow 2x &= \frac{1}{2}\ln\left(\frac{1+1/2}{1-1/2}\right) \\ \Rightarrow x &= \frac{1}{4}\ln 3.\end{aligned}$$

2 a)

$$\begin{aligned}y &= 4\sinh x - 6\cosh x, \\ \frac{dy}{dx} &= 4\cosh x - 6\sinh x.\end{aligned}$$

Turning points exist when $\frac{dy}{dx} = 0$:

$$\begin{aligned}4\cosh x - 6\sinh x &= 0 \\ \Rightarrow 4 - 6\tanh x &= 0\end{aligned}$$

$$\Rightarrow \tanh x = \frac{2}{3}$$
$$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+2/3}{1-2/3}\right) = \frac{1}{2}\ln 5.$$

$$\begin{aligned}
\frac{d^2y}{dx^2} &= 4\sinh x - 6\cosh x, \\
\left.\frac{d^2y}{dx^2}\right|_{x=\frac{1}{2}\ln 5} &= 4\sinh\left(\frac{1}{2}\ln 5\right) - 6\cosh\left(\frac{1}{2}\ln 5\right) \\
&= 4\frac{e^{\frac{1}{2}\ln 5} - e^{-\frac{1}{2}\ln 5}}{2} - 6\frac{e^{\frac{1}{2}\ln 5} + e^{-\frac{1}{2}\ln 5}}{2} \\
&= 2(5^{1/2} - 5^{-1/2}) - 3(5^{1/2} + 5^{-1/2}) \\
&= -5^{1/2} - 5\times 5^{-1/2} \\
&= -5^{1/2} - 5^{1/2} = -2\sqrt{5} < 0.
\end{aligned}$$

So the turning point is a local maximum.

2 b)

$$\begin{aligned}
V &= \pi\int_0^{\frac{1}{2}\ln 5} (4\sinh x - 6\cosh x)^2\,dx \\
&= \pi\int_0^{\frac{1}{2}\ln 5} (16\sinh^2 x - 48\sinh x\cosh x + 36\cosh^2 x)\,dx.
\end{aligned}$$

Using $\cosh 2x \equiv 2\cosh^2 x - 1$, $\cosh 2x \equiv 1 + 2\sinh^2 x$, and $\sinh 2x \equiv 2\sinh x\cosh x$:

$$\begin{aligned}
V &= \pi\int_0^{\frac{1}{2}\ln 5} \left(16\times\frac{1}{2}(\cosh 2x - 1) - 24\sinh 2x + 36\times\frac{1}{2}(\cosh 2x + 1)\right)dx \\
&= \pi\int_0^{\frac{1}{2}\ln 5} (26\cosh 2x - 24\sinh 2x + 10)\,dx \\
&= \pi[13\sinh 2x - 12\cosh 2x + 10x]_0^{\frac{1}{2}\ln 5} \\
&= \pi\left((13\sinh(2\times\frac{1}{2}\ln 5) - 12\cosh(2\times\frac{1}{2}\ln 5) + 10\times\frac{1}{2}\ln 5) - (13\sinh 0 - 12\cosh 0 + 10\times 0)\right) \\
&= \pi(13\sinh(\ln 5) - 12\cosh(\ln 5) + 5\ln 5 + 12) \\
&= \pi\left(13\frac{e^{\ln 5} - e^{-\ln 5}}{2} - 12\frac{e^{\ln 5} + e^{-\ln 5}}{2} + 5\ln 5 + 12\right) \\
&= \pi\left(13\frac{5 - 1/5}{2} - 12\frac{5 + 1/5}{2} + 5\ln 5 + 12\right) \\
&= \pi(12 + 5\ln 5).
\end{aligned}$$

So $p = 12$ and $q = 5$.

3)

$$\begin{aligned}
r &= 4\sqrt{\cosh 2\theta - \sinh 2\theta}, \\
A &= \frac{1}{2}\int_0^{\pi} r^2\,d\theta
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2}\int_0^{\pi} (4\sqrt{\cosh 2\theta - \sinh 2\theta})^2 \, d\theta \\
&= 8\int_0^{\pi} (\cosh 2\theta - \sinh 2\theta) \, d\theta \\
&= 8\left[\frac{1}{2}\sinh 2\theta - \frac{1}{2}\cosh 2\theta\right]_0^{\pi} \\
&= 8\left(\left(\frac{1}{2}\sinh 2\pi - \frac{1}{2}\cosh 2\pi\right) - \left(\frac{1}{2}\sinh 0 - \frac{1}{2}\cosh 0\right)\right) \\
&= 8\left(\frac{1}{2}\frac{e^{2\pi} - e^{-2\pi}}{2} - \frac{1}{2}\frac{e^{2\pi} + e^{-2\pi}}{2} + \frac{1}{2}\right) \\
&= 8\left(-\frac{1}{2}e^{-2\pi} + \frac{1}{2}\right) = 4 - 4e^{-2\pi}.
\end{aligned}
$$

4 a)

The point P is when $x = 0$:

$$
\begin{aligned}
0 &= 3\sinh t - \cosh t \\
\Rightarrow 0 &= 3\tanh t - 1 \\
\Rightarrow \tanh t &= \frac{1}{3} \\
\Rightarrow t &= \frac{1}{2}\ln\left(\frac{1 + 1/3}{1 - 1/3}\right) = \frac{1}{2}\ln 2.
\end{aligned}
$$

$$
\begin{aligned}
y\big|_{t=\frac{1}{2}\ln 2} &= \sinh\left(\frac{1}{2}\ln 2\right) + 2\cosh\left(\frac{1}{2}\ln 2\right) \\
&= \frac{e^{\frac{1}{2}\ln 2} - e^{-\frac{1}{2}\ln 2}}{2} + 2 \times \frac{e^{\frac{1}{2}\ln 2} + e^{-\frac{1}{2}\ln 2}}{2} \\
&= \frac{1}{2}\left(e^{\ln\sqrt{2}} - e^{\ln(\sqrt{2}/2)}\right) + e^{\ln\sqrt{2}} + e^{\ln(\sqrt{2}/2)} \\
&= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{4} + \sqrt{2} + \frac{\sqrt{2}}{2} = \frac{7\sqrt{2}}{4}.
\end{aligned}
$$

So P has coordinates $(0, \frac{7\sqrt{2}}{4})$.

4 b)

$$
\begin{aligned}
x\big|_{t=0} &= 3\sinh 0 - \cosh 0 = -1, \\
y\big|_{t=0} &= \sinh 0 + 2\cosh 0 = 2.
\end{aligned}
$$

So Q has coordinates $(-1, 2)$.

$$
\begin{aligned}
\frac{dx}{dt} &= 3\cosh t - \sinh t, \\
\frac{dy}{dt} &= \cosh t + 2\sinh t, \\
\frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{\cosh t + 2\sinh t}{3\cosh t - \sinh t}, \\
\left.\frac{dy}{dx}\right|_{t=0} &= \frac{\cosh 0 + 2\sinh 0}{3\cosh 0 - \sinh 0} = \frac{1}{3}.
\end{aligned}
$$

So the tangent has equation $y - 2 = \frac{1}{3}(x + 1)$, hence $3y - 6 = x + 1$ and $-x + 3y = 7$.

5)

$$\begin{aligned} x &= 5\sinh t - 8\cosh t, \\ y &= 2\cosh t - 3\sinh t, \\ \therefore x + y &= 2\sinh t - 6\cosh t. \end{aligned}$$

The maximum value of $x + y$ occurs when

$$\begin{aligned} \frac{d}{dt}(2\sinh t - 6\cosh t) &= 0, \\ 2\cosh t - 6\sinh t &= 0 \\ \Rightarrow 1 - 3\tanh t &= 0 \\ \Rightarrow \tanh t &= \frac{1}{3} \\ \Rightarrow t &= \operatorname{artanh}\frac{1}{3} = \frac{1}{2}\ln\left(\frac{1+1/3}{1-1/3}\right) = \frac{1}{2}\ln 2. \end{aligned}$$

$$\begin{aligned} \max(x+y) &= 2\sinh\left(\frac{1}{2}\ln 2\right) - 6\cosh\left(\frac{1}{2}\ln 2\right) \\ &= 2\frac{e^{\frac{1}{2}\ln 2} - e^{-\frac{1}{2}\ln 2}}{2} - 6\frac{e^{\frac{1}{2}\ln 2} + e^{-\frac{1}{2}\ln 2}}{2} \\ &= e^{\frac{1}{2}\ln 2} - e^{-\frac{1}{2}\ln 2} - 3(e^{\frac{1}{2}\ln 2} + e^{-\frac{1}{2}\ln 2}) \\ &= -2e^{\frac{1}{2}\ln 2} - 4e^{-\frac{1}{2}\ln 2} \\ &= -2e^{\ln\sqrt{2}} - 4e^{\ln(\sqrt{2}/2)} \\ &= -2\sqrt{2} - 4\frac{\sqrt{2}}{2} = -4\sqrt{2}. \end{aligned}$$

Therefore $x + y = -4\sqrt{2}$.

6 a)

$$\begin{aligned} x^2 + 3y^2 &= (2\sinh 2t - \cosh 2t)^2 + 3(\sinh 2t)^2 \\ &= 4\sinh^2 2t - 4\sinh 2t\cosh 2t + \cosh^2 2t + 3\sinh^2 2t \\ &= 7\sinh^2 2t - 4\sinh 2t\cosh 2t + \cosh^2 2t \\ &= 7\sinh^2 2t - 4\sinh 2t\cosh 2t + (1 + \sinh^2 2t) \\ &= 8\sinh^2 2t - 4\sinh 2t\cosh 2t + 1 \\ &= 4(2\sinh 2t - \cosh 2t)\sinh 2t + 1 \\ &= 4xy + 1, \quad \text{as required.} \end{aligned}$$

6 b)

$$A = \int_0^{\ln 2} y\frac{dx}{dt}\,dt,$$

$$y = \sinh 2t, \quad x = 2\sinh 2t - \cosh 2t, \quad \frac{dx}{dt} = 4\cosh 2t - 2\sinh 2t,$$

$$\therefore A = \int_0^{\ln 2} \sinh 2t(4\cosh 2t - 2\sinh 2t)\,dt$$
$$= \int_0^{\ln 2} (4\sinh 2t\cosh 2t - 2\sinh^2 2t)\,dt.$$

Note that $\sinh 2A \equiv 2\sinh A\cosh A$ and $\cosh 2A \equiv 1 + 2\sinh^2 A$. Thus $\sinh 4t \equiv 2\sinh 2t\cosh 2t$ and $\cosh 4t = 1 + 2\sinh^2 2t$.

$$\therefore A = \int_0^{\ln 2} (2\sinh 4t - \cosh 4t + 1)\,dt$$
$$= \left[\frac{1}{2}\cosh 4t - \frac{1}{4}\sinh 4t + t\right]_0^{\ln 2}$$
$$= \left(\frac{1}{2}\frac{e^{4\ln 2} + e^{-4\ln 2}}{2} - \frac{1}{4}\frac{e^{4\ln 2} - e^{-4\ln 2}}{2} + \ln 2\right) - \left(\frac{1}{2} - 0 + 0\right)$$
$$= \frac{1}{4}(16 + 1/16) - \frac{1}{8}(16 - 1/16) + \ln 2 - \frac{1}{2}$$
$$= \frac{257}{64} - \frac{255}{128} + \ln 2 - \frac{1}{2} = \frac{195}{128} + \ln 2.$$

7 a)

$$\cosh(iz) = \frac{e^{iz} + e^{-iz}}{2}$$
$$= \frac{(\cos z + i\sin z) + (\cos z - i\sin z)}{2}$$
$$= \frac{2\cos z}{2} = \cos z.$$

7 b)

$$\cos z = 4 \Rightarrow \cosh(iz) = 4$$
$$\Rightarrow iz = \pm\operatorname{arcosh} 4$$
$$\Rightarrow iz = \pm\ln(4 + \sqrt{4^2 - 1})$$
$$\Rightarrow iz = \pm\ln(4 + \sqrt{15})$$
$$\Rightarrow z = \pm\frac{\ln(4 + \sqrt{15})}{i}$$
$$\Rightarrow z = \pm i\ln(4 + \sqrt{15}).$$

Or $z = i\ln(4 \pm \sqrt{15})$.

8 a)

$$3\sinh x + 5\cosh x \equiv 3\frac{e^x - e^{-x}}{2} + 5\frac{e^x + e^{-x}}{2}$$
$$\equiv \frac{3}{2}(e^x - e^{-x}) + \frac{5}{2}(e^x + e^{-x})$$
$$\equiv \frac{3}{2}e^x - \frac{3}{2}e^{-x} + \frac{5}{2}e^x + \frac{5}{2}e^{-x}$$

$$\equiv 4e^x + e^{-x}$$
$$\equiv e^{-x}(1 + 4e^{2x}).$$

8 b)

$$\int_{\ln(\sqrt{3}/6)}^{\ln(1/2)} \frac{1}{3\sinh x + 5\cosh x}\,dx = \int_{\ln(\sqrt{3}/6)}^{\ln(1/2)} \frac{1}{e^{-x}(1+4e^{2x})}\,dx$$
$$= \int_{\ln(\sqrt{3}/6)}^{\ln(1/2)} \frac{e^x}{1+4e^{2x}}\,dx.$$

Notice that the denominator is $1 + u^2$ if $u = 2e^x$. Then $du/dx = 2e^x$, so $dx = \frac{1}{2e^x}du$.

$$x = \ln\frac{1}{2} \Rightarrow u = 2e^{\ln(1/2)} = 1, \qquad x = \ln\frac{\sqrt{3}}{6} \Rightarrow u = 2e^{\ln(\sqrt{3}/6)} = \frac{\sqrt{3}}{3}.$$

$$\therefore \int \frac{e^x}{1+4e^{2x}}\,dx = \int_{\sqrt{3}/3}^{1} \frac{e^x}{1+u^2} \times \frac{1}{2e^x}\,du$$
$$= \frac{1}{2}\int_{\sqrt{3}/3}^{1} \frac{1}{1+u^2}\,du$$
$$= \frac{1}{2}[\arctan u]_{\sqrt{3}/3}^{1}$$
$$= \frac{1}{2}\left(\arctan 1 - \arctan\frac{\sqrt{3}}{3}\right)$$
$$= \frac{1}{2}\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \frac{\pi}{24}.$$

9 a)

$$f(0) = \operatorname{artanh}\left(\frac{2\times 0 - 1}{2 - 2\times 0}\right)$$
$$= \operatorname{artanh}\left(-\frac{1}{2}\right)$$
$$= \frac{1}{2}\ln\left(\frac{1+(-1/2)}{1-(-1/2)}\right)$$
$$= \frac{1}{2}\ln\frac{1}{3} = \ln\left(\left(\frac{1}{3}\right)^{1/2}\right)$$
$$= \ln\frac{1}{\sqrt{3}} = \ln\frac{\sqrt{3}}{3}.$$

9 b)

$$f(x) = \operatorname{artanh}\left(\frac{2x-1}{2-2x}\right),$$
$$f'(x) = \frac{(2-2x)\times 2 - (2x-1)\times(-2)}{(2-2x)^2} \times \frac{1}{1 - \left(\frac{2x-1}{2-2x}\right)^2}$$

$$
\begin{aligned}
&= \frac{2}{(2-2x)^2} \times \frac{1}{1-\frac{(2x-1)^2}{(2-2x)^2}} \\
&= \frac{2}{(2-2x)^2} \times \frac{(2-2x)^2}{(2-2x)^2-(2x-1)^2} \\
&= \frac{2}{1} \times \frac{1}{4-8x+4x^2-(4x^2-4x+1)} \\
&= \frac{2}{3-4x}.
\end{aligned}
$$

9 c)

$$
\begin{aligned}
f''(x) &= \frac{(3-4x)\times 0 - 2\times(-4)}{(3-4x)^2} \\
&= \frac{8}{(3-4x)^2}.
\end{aligned}
$$

9 d)

$$
\begin{aligned}
f(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots, \\
f(0) &= \ln\frac{\sqrt{3}}{3}, \\
f'(0) &= \frac{2}{3-4\times 0} = \frac{2}{3}, \\
f''(0) &= \frac{8}{(3-4\times 0)^2} = \frac{8}{9}, \\
\therefore f(x) &= \ln\frac{\sqrt{3}}{3} + \frac{2}{3}x + \frac{8/9}{2!}x^2 + \cdots \\
&\approx \ln\frac{\sqrt{3}}{3} + \frac{2}{3}x + \frac{4}{9}x^2.
\end{aligned}
$$

10)

$$
\begin{aligned}
f(x) = \sinh ix &= \frac{e^{ix}-e^{-ix}}{2} \\
&= \frac{(\cos x + i\sin x) - (\cos x - i\sin x)}{2} \\
&= \frac{2i\sin x}{2} = i\sin x, \\
y = (f(x))^4 &= (i\sin x)^4 = \sin^4 x.
\end{aligned}
$$

Mean value:

$$\frac{1}{2\pi - 0}\int_0^{2\pi} \sin^4 x\, dx.$$

Now $(\cos x + i\sin x)^4 = \cos 4x + i\sin 4x$ and $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$, so

$$\sin^4 x = \left(\frac{e^{ix}-e^{-ix}}{2i}\right)^4$$

$$
\begin{aligned}
&= \frac{(e^{ix} - e^{-ix})^4}{16} \\
&= \frac{e^{4ix} - 4e^{2ix} + 6 - 4e^{-2ix} + e^{-4ix}}{16} \\
&= \frac{e^{4ix} + e^{-4ix} - 4(e^{2ix} + e^{-2ix}) + 6}{16}.
\end{aligned}
$$

Meanwhile, $\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} \Rightarrow e^{i\theta} + e^{-i\theta} = 2\cos\theta$. Therefore

$$
\sin^4 x = \frac{2\cos 4x - 8\cos 2x + 6}{16} = \frac{1}{8}\cos 4x - \frac{1}{2}\cos 2x + \frac{3}{8}.
$$

So the mean value is

$$
\begin{aligned}
&= \frac{1}{2\pi}\int_0^{2\pi} \left(\frac{1}{8}\cos 4x - \frac{1}{2}\cos 2x + \frac{3}{8}\right) dx \\
&= \frac{1}{2\pi}\left[\frac{1}{32}\sin 4x - \frac{1}{4}\sin 2x + \frac{3}{8}x\right]_0^{2\pi} \\
&= \frac{1}{2\pi}\left((0 - 0 + \frac{3\pi}{4}) - (0 - 0 + 0)\right) \\
&= \frac{1}{2\pi} \times \frac{3\pi}{4} = \frac{3}{8}.
\end{aligned}
$$

Alternatively, $f(x) = i\sin x$ and $y = (f(x))^4 = \sin^4 x$. The mean value is again $\frac{1}{2\pi}\int_0^{2\pi} \sin^4 x\, dx$. Since

$$
\begin{aligned}
\sin^4 x = (\sin^2 x)^2 &= \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)^2 \\
&= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{4}\cos^2 2x \\
&= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}\cos 4x\right) \\
&= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{8} + \frac{1}{8}\cos 4x \\
&= \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x,
\end{aligned}
$$

the mean value is

$$
\begin{aligned}
&= \frac{1}{2\pi}\int_0^{2\pi} \left(\frac{1}{8}\cos 4x - \frac{1}{2}\cos 2x + \frac{3}{8}\right) dx \\
&= \frac{1}{2\pi}\left[\frac{1}{32}\sin 4x - \frac{1}{4}\sin 2x + \frac{3}{8}x\right]_0^{2\pi} \\
&= \frac{1}{2\pi} \times \frac{3\pi}{4} = \frac{3}{8}.
\end{aligned}
$$

11 a)

$$
\begin{aligned}
\cosh((p+1)x) &= \cosh(px + x) = \cosh px \cosh x + \sinh px \sinh x, \\
\cosh((p-1)x) &= \cosh(px - x) = \cosh px \cosh(-x) + \sinh px \sinh(-x) \\
&= \cosh px \cosh x - \sinh px \sinh x.
\end{aligned}
$$

11 b)

$$\begin{aligned}\cosh((p+1)x) - \cosh((p-1)x) &= (\cosh px \cosh x + \sinh px \sinh x) \\ &\quad - (\cosh px \cosh x - \sinh px \sinh x) \\ &= 2 \sinh px \sinh x.\end{aligned}$$

So $\cosh((p+1)x) - \cosh((p-1)x) = 2 \sinh px \sinh x$, and hence

$$\sinh px = \frac{\cosh((p+1)x) - \cosh((p-1)x)}{2 \sinh x}.$$

11 c)

$$\begin{aligned}S_n &= \sinh x + \sinh 2x + \cdots + \sinh nx \\ &= \sum_{k=1}^{n} \frac{\cosh((k+1)x) - \cosh((k-1)x)}{2 \sinh x}.\end{aligned}$$

$$\begin{aligned}k = 1 : &\quad \frac{\cosh 2x - \cosh 0}{2 \sinh x} \\ k = 2 : &\quad \frac{\cosh 3x - \cosh x}{2 \sinh x} \\ k = 3 : &\quad \frac{\cosh 4x - \cosh 2x}{2 \sinh x} \\ &\vdots \\ k = n-2 : &\quad \frac{\cosh((n-1)x) - \cosh((n-3)x)}{2 \sinh x} \\ k = n-1 : &\quad \frac{\cosh nx - \cosh((n-2)x)}{2 \sinh x} \\ k = n : &\quad \frac{\cosh((n+1)x) - \cosh((n-1)x)}{2 \sinh x}\end{aligned}$$

Therefore

$$\begin{aligned}S_n &= \frac{\cosh nx}{2 \sinh x} + \frac{\cosh((n+1)x)}{2 \sinh x} - \frac{\cosh 0}{2 \sinh x} - \frac{\cosh x}{2 \sinh x} \\ &= \frac{\cosh nx + \cosh((n+1)x) - \cosh x - 1}{2 \sinh x}.\end{aligned}$$

12)

$$\frac{dy}{dx} + y \coth x = \cosh^2 x,$$

Integrating Factor: $e^{\int \coth x \, dx} = e^{\ln|\sinh x|} = \sinh x.$

$$\begin{aligned}\Rightarrow \sinh x \frac{dy}{dx} + y \sinh x \coth x &= \sinh x \cosh^2 x \\ \Rightarrow \sinh x \frac{dy}{dx} + y \cosh x &= \sinh x \cosh^2 x \\ \Rightarrow \frac{d}{dx}(y \sinh x) &= \sinh x \cosh^2 x\end{aligned}$$

$$\Rightarrow y\sinh x = \frac{1}{3}\cosh^3 x + c.$$

When $y = 2$ and $x = \ln 2$:

$$\begin{aligned} 2\sinh(\ln 2) &= \frac{1}{3}\cosh^3(\ln 2) + c \\ \Rightarrow 2 \times \frac{e^{\ln 2} - e^{-\ln 2}}{2} &= \frac{1}{3}\left(\frac{e^{\ln 2} + e^{-\ln 2}}{2}\right)^3 + c \\ \Rightarrow \frac{3}{2} &= \frac{125}{192} + c \\ \Rightarrow c &= \frac{163}{192}. \end{aligned}$$

Therefore

$$y\sinh x = \frac{1}{3}\cosh^3 x + \frac{163}{192}, \qquad y = \frac{\cosh^3 x}{3\sinh x} + \frac{163}{192\sinh x}.$$

13)

$$\frac{dy}{dx} - y\tanh x = \sinh 2x,$$

Integrating Factor: $e^{\int -\tanh x\,dx} = e^{-\ln|\cosh x|} = e^{\ln|1/\cosh x|} = e^{\ln|\operatorname{sech} x|}$.

$$\begin{aligned} \Rightarrow \operatorname{sech} x\frac{dy}{dx} - y\operatorname{sech} x\tanh x &= \operatorname{sech} x\sinh 2x \\ \Rightarrow \operatorname{sech} x\frac{dy}{dx} - y\operatorname{sech} x\tanh x &= 2\sinh x \\ \Rightarrow \frac{d}{dx}(y\operatorname{sech} x) &= 2\sinh x \\ \Rightarrow y\operatorname{sech} x &= 2\cosh x + c. \end{aligned}$$

When $y = 5$ and $x = 0$:

$$5\operatorname{sech} 0 = 2\cosh 0 + c \quad \Rightarrow \quad 5 = 2 + c \quad \Rightarrow \quad c = 3.$$

Therefore $y\operatorname{sech} x = 2\cosh x + 3$, and

$$y = 2\cosh^2 x + 3\cosh x.$$